

Model Evaluation Toolkit

A streamlined Python toolkit for comprehensive model evaluation. `met-gfit` automates the calculation of 10 key statistical metrics and generates professional diagnostic plots (1:1 scatter, Residual, and Q-Q plots) to validate and visualize model accuracy with ease.

Input

- At minimum, you need to provide two vectors:
 - `true`: observed (actual) values
 - `pred`: predicted values
- Optional inputs allow you to control the number of decimal places for statistics and whether to generate plots.

Comprehensive Evaluation

The core function `gfit()` performs a full-scale diagnostic of your model, providing:

- **10 Statistical Metrics:** Returns a dictionary/report containing R^2 , $RMSE$, MAE , $Bias$, r , d -index, and more.
- **Visual Diagnostics:** Optionally generates a professional three-panel figure including:
 - **1:1 Scatter Plot:** Visualizes accuracy and potential scaling issues.
 - **Residual Plot:** Checks for non-linearity and heteroscedasticity.
 - **Q-Q Plot:** Evaluates the normality of the error distribution.

File Structure

The repository is organized following the standard Python `src` layout for better compatibility and installation:

- `src/met_gfit/`:
 - `metrics.py`: The engine containing all mathematical formulas and plotting logic.
 - `__init__.py`: The gateway that allows for clean imports (e.g., `from met_gfit import gfit`).
- `pyproject.toml`: Project metadata and dependency specifications.
- [README.md](#): This comprehensive documentation and user guide.
- `LICENSE`: MIT License for open-source use.

Considered goodness-of-fit statistics

10 statistics are included in this function "[gfit.py](#)". They are:

- **MSE: Mean Squared Error.** Square of the unit of the output variable.
Measures the **average squared difference** between observed and predicted values. Lower values indicate a better fit.
- **RMSE: Root Mean Square Error.** The unit is the same as output variable.
The square root of MSE; shows the **average error** in the same units as the data.
- **Bias: Bias.** The unit is the same as output variable.
The average difference between predicted and observed values. Indicates if predictions are systematically too high or too low.
- **r: Correlation coefficient.** Unitless.
Measures the strength and direction of the linear relationship between observed and predicted values (ranges from -1 to 1).
- **p-value: p-value.** Unitless.
 - It tells how likely it is that the result happened by random chance.
 - A **small p-value** (usually less than 0.05) means the result is probably not just random, so there is likely a real effect of the relationship.
 - A **large p-value** means the result could easily happen by chance, so there may not be a real effect.
- **d: Willmoot's index of agreement.** Unitless.
Measures the degree of model prediction error (ranges from 0 to 1, with 1 being perfect agreement).
- **R2: Coefficient of determination.** Unitless.
Shows the proportion of variance in observed values explained by the predictions (ranges from 0 to 1).
- **MAE: Mean Absolute Error.** The unit is the same as the output variable.
The average of the absolute differences between observed and predicted values.
- **RRMSE: Relative Root Mean Square Error.** %.
RMSE expressed as a percentage of the mean observed value.
- **RSD: The standard deviation of the residual.** The unit is the same as the output variable.
Indicates how spread out the residuals (errors)

Equations

met_gfit provides a comprehensive assessment of model performance, covering accuracy, precision, and agreement with observations. The following statistics are implemented:

Mean Square Error (*MSE*)

Measures the average of the squares of the errors, giving more weight to larger deviations (outliers).

$$MSE = \frac{1}{n} \sum_{i=1}^n (o_i - p_i)^2$$

Root Mean Square Error ($RMSE$)

Indicates how much the model's predictions deviate from the observations on average.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (o_i - \hat{p}_i)^2}{n}}$$

Mean Bias ($Bias$)

Measures the average tendency of the model to over- or under-estimate the observed values. A positive value indicates overestimation.

$$Bias = \frac{1}{n} \sum_{i=1}^n (p_i - o_i)$$

Pearson Correlation Coefficient (r)

Indicates the strength and direction of the linear relationship between predicted and observed values. A correlation close to 1 or -1 indicates that the model captures the variability of observations well.

$$r = \frac{cov(o, p)}{\sigma_o \sigma_p}$$

Willmott's Index of Agreement (d)

Considers both the mean absolute error and the variability of the observations, showing insight into how well the model replicates the variability and pattern of the observations (Willmott, 1982).

$$d = 1 - \frac{\sum_{i=1}^n (o_i - p_i)^2}{\sum_{i=1}^n (|p_i - \bar{o}| + |o_i - \bar{o}|)^2}$$

Coefficient of Determination (R^2)

Quantifies the proportion of the variance in the observations that is predictable from the model, providing a rigorous measure of model fit.

$$R^2 = 1 - \frac{\sum_{i=1}^n (o_i - \hat{p}_i)^2}{\sum_{i=1}^n (o_i - \bar{o})^2}$$

Mean Absolute Error (*MAE*)

Represents the average of the absolute differences between observations and predictions, providing a linear score where all individual differences are weighted equally.

$$MAE = \frac{1}{n} \sum_{i=1}^n |o_i - p_i|$$

Relative Root Mean Square Error (*RRMSE*)

Normalizes the *RMSE* by the mean of the observations, expressed as a percentage, which allows for comparison between different datasets or variables.

$$RRMSE = \frac{RMSE}{\bar{o}} \times 100\%$$

Residual Standard Deviation (*RSD*)

Quantifies the spread of the residuals (the differences between model and observations) around their mean.

$$RSD = \sqrt{\frac{\sum_{i=1}^n (p_i - o_i - \frac{1}{n} \sum_{i=1}^n (p_i - o_i))^2}{n - 1}}$$

where *o* represents observations, *o_i* is the observed value for the *i*th observations, $\bar{o}$ is the mean of observations, *p* represents predictions, *p_i* is the estimated value for the *i*th estimations, $\hat{p}_i$ is the estimated value corresponding to the *i*th observations, *n* is the number of paired observations.

Reference

- Willmott, C. J. (1982). Some comments on the evaluation of model performance. Bulletin of the American Meteorological Society, 63(11), 1309-1313.
[https://doi.org/10.1175/1520-0477\(1982\)063<1309:SCOTEO>2.0.CO;2](https://doi.org/10.1175/1520-0477(1982)063<1309:SCOTEO>2.0.CO;2)
- Nash, J. E., & Sutcliffe, J. V. (1970). River flow forecasting through conceptual models part I — A discussion of principles. Journal of Hydrology, 10(3), 282-290. (For *NSE/R²* implementation).

Software Citation

If you also use this repository in your work, please cite it using the following DOI:
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